

Notes on Approximation Theory

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The notes in this page catalog results that may be useful in finding bounds on how close a polynomial or rational function is to a single-variable function on a compact interval.

The aim is to find error bounds that are *explicit*, with no hidden constants and without introducing transcendental or trigonometric functions. If an error bound is explicit, it can be computed *offline*, without performing an approximation first, so that it can be known, for example, which degree polynomial to build in order to come close to a function with a given accuracy.

The mapping from a function to a function (for example, from a single-variable function to a polynomial “close” to it) is called an *operator*, and operators involved in these bounds are often linear operators, whose behavior is relatively simple to examine.

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2 About This Document

This is an open-source document; for an updated version, see the source code¹ or its rendering on GitHub². You can send comments on this document on the GitHub issues page³, especially if you find any errors on this page.

3 Notation and Definitions

For definitions of *continuous*, *derivative*, *convex*, *concave*, *Hölder continuous*, and *Lipschitz continuous*, see the definitions section in “**Supplemental Notes for Bernoulli Factory Algorithms**”⁴.

- The *closed unit interval* (written as $[0, 1]$) means the set consisting of 0, 1, and every real number in between.
- A *compact interval* (written as $[a, b]$), means a set of real numbers consisting of a , b , and every real number in between, where a is less than or equal to b .⁵ The closed unit interval is one example of a compact interval.
- An *operator* is a mapping from a function to a function.
- An operator L is *linear* if it satisfies $L(af) = aL(f)$ and $L(f+g) = L(f) + L(g)$ for all allowed functions f and g and every number a .
- An operator L is *positive* if it has the property that, if an allowed function f is nonnegative on its domain, so is $L(f)$.⁶
- The *operator norm* of an operator L is the maximum absolute value of $L(f)$ over all allowed functions f with a maximum absolute value 1 or less. This assumes L maps continuous functions on a compact interval to functions of that kind.
- An operator is *bounded* if its operator norm is finite. A function $f(\lambda)$ is *bounded* if there are two real numbers a and b such that $a < f(\lambda) < b$ over the domain of f .
- The *expected value* (or mean or “long-run average”) of a random variable Y is denoted $\mathbb{E}[Y]$.
- A *modulus of continuity of order 1* of a function f , denoted $\omega_1(f, \delta)$, means a nonnegative and nowhere decreasing function where, for each $\delta \geq 0$, $\text{abs}(f(x) - f(y)) \leq \omega_1(f, \delta)$ whenever x and y are in f ’s domain and no more than δ apart. Loosely speaking, $\omega_1(f, \delta)$ gives how much f can vary when f is restricted to a window of size δ or less. The modulus of continuity reflects the “regularity” of f ; generally, the smaller it is, the more “regular”.
- In this document:
 - e_i is a function such that $e_i(t) = t^i$, so that $e_0(t) = 1$ and $e_1(t) = t$; as an example, if $L(f) = f(0) + f(1)$, then $L(e_1 - x) = (e_1(0) - x) + (e_1(1) - x) = (0 - x) + (1 - x) = 1 - 2x$.
 - The notation $\|f\|$, where f is a function, means the maximum absolute value over the function’s domain unless noted otherwise (for example, where the notation means an operator norm instead).

4 Bernstein Form and Bernstein Polynomials

Among the best known examples of linear operators are the Bernstein polynomials.

In this document, a polynomial $P(x)$ is written in *Bernstein form of degree n* if it is written as—

¹<https://github.com/peteroupc/peteroupc.github.io/raw/master/approxtheory.md>

²<https://github.com/peteroupc/peteroupc.github.io/blob/master/approxtheory.md>

³<https://github.com/peteroupc/peteroupc.github.io/issues>

⁴<https://peteroupc.github.io/bernsupp.html#Definitions>

⁵This term is used instead of “closed interval”, a term which can also encompass infinitely long intervals, which is not the intent of this document (Weisstein, Eric W. “Closed Interval.” From MathWorld—A Wolfram Resource. <https://mathworld.wolfram.com/ClosedInterval.html>).

⁶A better term for positive operators is probably nonnegativity-preserving operators.

$$P(x) = \sum_{k=0}^n a_k \frac{n!}{(k!)((n-k)!)} x^k (1-x)^{n-k},$$

where $0 \leq x \leq 1$ and the real numbers $a_0, \dots, a_n$ are the polynomial's *Bernstein coefficients*.⁷

The degree- n *Bernstein polynomial* of an arbitrary function $f(x)$ has Bernstein coefficients $a_k = f(k/n)$. In general, this Bernstein polynomial differs from f even if f is a polynomial. In this document, the degree- n Bernstein polynomial of f is denoted $B_n(f)$.

$B_n(f)$ is a positive linear operator. It maps a bounded function on the closed unit interval to a polynomial of degree n or less on that interval.

This page's emphasis is on methods that produce polynomials in Bernstein form, ratios of such polynomials, or functions like $f(g(x))$ where f and g are such polynomials or ratios.

5 “Moments” of Linear Operators

To examine the approximation behavior of linear operators, it is helpful to find the so-called “moments” of those operators, that is, the functions they map certain functions to.

For a linear operator L , they are:

- “Raw moments”: The values of $L(e_i)$ for each integer $i \geq 0$.
- “Central moments”: The values of $L((e_1 - x)^i)$ for each integer $i \geq 0$. If the “raw moments” $L(e_0), \dots, L(e_j)$ are known, then $L((e_1 - x)^j)$ is also known, thanks to proposition 5.6 of Gonska et al. (2006)⁸.
- “Absolute moments”: The values of $L(\text{abs}(e_1 - x)^i)(x)$ for each integer $i \geq 0$. When i is even, $L(\text{abs}(e_1 - x)^i) = L((e_1 - x)^i)$.

Because L is linear, if $L(e_i) = e_i$ for each i from 0 through j (j is zero or a positive integer), then:

- L reproduces all polynomials up to degree j (that is, $L(f) = f$ whenever f is a polynomial of degree j or less).
- The 0-th “central moment” is $L(e_0) = L((e_1 - x)^0) = 1$, and for each i from 1 through j , $L((e_1 - x)^j) = 0$.

Also, because L is linear, the “moments” of degree up to m , say, lead to easy ways to find the mapping by L of any polynomial of degree up to m , when the polynomial is written in “power” form.

Example: Let $f(x)$ be the polynomial $4x^3 - 6x^2 + 8x^1 - 10$. Then:

$$L(f) = 4L(e_3) - 6L(e_2) + 8L(e_1) - 10L(e_0).$$

5.1 “Moments” of Bernstein Polynomials

The following results deal with useful quantities when discussing the error in approximating a function by Bernstein polynomials.

Suppose a coin shows heads with probability p , and n independent tosses of the coin are made, where n is 1 or greater. Then the total number of heads X follows a *binomial distribution*. The following are useful quantities of this distribution.

⁷ $n! = 1*2*3*\dots*n$ is also known as n factorial; in this document, $(0!) = 1$. *Summation notation*, involving the Greek capital sigma (Σ), is a way to write the sum of one or more terms of similar form. For example, $\sum_{k=0}^n g(k)$ means $g(0) + g(1) + \dots + g(n)$, and $\sum_{k \geq 0} g(k)$ means $g(0) + g(1) + \dots$.

⁸Gonska, Heiner, Paula Pi ul, and Ioan Raşa. “On differences of positive linear operators.” *Carpathian Journal of Mathematics* (2006): 65-78.

- $T_{n,r}$: The *central moment* (moment about the mean) of X is denoted $T_{n,r}(p) = \mathbb{E}[(X - \mathbb{E}[X])^r] = B_n((e_1 - p)^r)(p) \cdot n^r$. Formulas for computing this central moment are given in Skorski (2024)⁹.
- $S_{n,r}$: Traditionally, the central moment of X/n or the ratio of heads to tosses is denoted $S_{n,r}(p) = T_{n,r}(p)/n^r = \mathbb{E}[(X/n - \mathbb{E}[X/n])^r] = B_n((e_1 - p)^r)(p)$. (T and S are notations of S.N. Bernstein, known for Bernstein polynomials.) $S_{n,r}$ is thus the r -th “central moment” of degree- n Bernstein polynomials.
- $M_{n,r}$: The r -th *central absolute moment* of X/n is denoted $M_{n,r}(p) = \mathbb{E}[\text{abs}(X/n - \mathbb{E}[X/n])^r] = B_n(\text{abs}(e_1 - p)^r)(p)$. If r is even, $M_{n,r}(p) = S_{n,r}(p)$. $M_{n,r}$ is thus the r -th “absolute moment” of degree- n Bernstein polynomials.

The following gives bounds on $M_{n,r}$; some results in approximation theory rely on bounds like these.¹⁰

Proposition 1: *Let $r \geq 0$, and let $\sigma(r, t) = (r!)/(((r/2)!)t^{r/2})$. Then for real numbers r and integers n described in the following table, $M_{n,r}(p) \leq \mu_{n,r}/n^{r/2}$, where $\mu_{n,r}$ is as given in the table.*

If $r \dots$	Then $\mu_{n,r}$ is...
Is an even integer.	$\sigma(r, 6)$, for positive n .
Is an even integer, but not greater than 44.	$\sigma(r, 8)$, for positive n .
Is 1.	$1/2$, for positive n .
Is odd, and $3 \leq r \leq 43$.	$\sqrt{\sigma(r-1, 8)\sigma(r+1, 8)} = r^{1/2}(r-1)!/(2 \cdot 8^{(r-1)/2}((r-1)/2)!)$, for $n \geq 2$.
Is odd and greater than 43.	$\sqrt{\sigma(r-1, 6)\sigma(r+1, 6)}$, for $n \geq 2$.

Proof: The first row comes from a result of Adell and Cárdenas-Morales (2018)¹¹. The second row is an improved result of the first, from Molteni (2022)¹². The third row follows from Cheng (1983)¹³. The fourth and fifth rows follow from the first and second as well as that the absolute central moment for odd r can be bounded for every integer $n \geq 2$, using Schwarz’s inequality (Weisstein)¹⁴ (see also Bojanić and Shisha 1975¹⁵ for the case $r = 4$). []

6 Taylor Expansion of Linear Operators

Continuous functions can be “unwrapped” into a Taylor expansion. The linear mapping of those functions also has a Taylor expansion of sorts, which is described next.

Let $f(\lambda)$ have a continuous s -th derivative on a compact interval, where s is zero or a positive integer, and let $L(f)$ be a linear operator that maps continuous functions on that interval to functions of that kind. Then:

$$L(f)(\lambda) = L(R_s(f, \lambda)) + \sum_{i=0}^s L((e_1 - \lambda)^i)(\lambda) \frac{f^{(i)}(\lambda)}{i!}, \quad (1)$$

⁹Skorski, Maciej. “Handy formulas for binomial moments.” *Modern Stochastics: Theory and Applications* 12.1 (2024): 27-41.

¹⁰It is also possible to bound the “absolute moment” as $M_{n,r}(p) \leq C(r)(\max(1/n, (p(1-p)/n)^{1/2}))^r$ or $M_{n,r}(p) \leq D(r)(1/n + (p(1-p)/n)^{1/2})^r$ (G.G. Lorentz, “The degree of approximation by polynomials with positive coefficients”, 1966), but the constants $C(r)$ and $D(r)$ seem to be higher (and less favorable) than the $E(r)$ in $M_{n,r}(p) \leq E(r)/n^{r/2}$.

¹¹Adell, J.A., Cárdenas-Morales, D., “Quantitative generalized Voronovskaja’s formulae for Bernstein polynomials”, *Journal of Approximation Theory* 231, July 2018. <https://doi.org/10.1016/j.jat.2018.04.007> <https://www.sciencedirect.com/science/article/pii/S0021904518300376>

¹²Molteni, Giuseppe. “Explicit bounds for even moments of Bernstein’s polynomials.” *Journal of Approximation Theory* 273 (2022): 105658.

¹³Cheng, F., “On the rate of convergence of Bernstein polynomials of functions of bounded variation”, *Journal of Approximation Theory* 39 (1983).

¹⁴Weisstein, Eric W. “Schwarz’s Inequality.” From MathWorld—A Wolfram Resource. <https://mathworld.wolfram.com/SchwarzsInequality.html>

¹⁵R. Bojanić, O. Shisha, “Degree of L^1 approximation to integrable functions by modified Bernstein polynomials”, *Journal of Approximation Theory* 13, 66–72 (1975).

where $R_s(f, \lambda)$ is the remainder after subtracting from f the degree- s Taylor polynomial of f centered at λ . (See also Pițul (2007, proof of theorem 5.8)¹⁶.) $R_s(f, \lambda)$ is 0 if f is a polynomial of degree s or less.

If L reproduces constants, so that $L(e_0) = 1$, this becomes:

$$L(f)(\lambda) - f(\lambda) = L(R_s(f, \lambda)) + \sum_{i=1}^s L((e_1 - \lambda)^i)(\lambda) \frac{f^{(i)}(\lambda)}{i!}. \quad (2)$$

If L reproduces polynomials up to degree s , this even reduces to $L(f)(\lambda) - f(\lambda) = L(R_s(f, \lambda))$.

It can be seen from the expansions just given that finding upper bounds for $L_n(f)(\lambda)$ involves:

- Finding upper bounds for L 's “central moments” up to the s -th order.
- Finding upper bounds for $L(R_s(f, \lambda))$. If L is positive linear, such bounds are given in the section “**Bounds for General Positive Linear Operators**”. If L is nonpositive linear, bounds are given in the section “**Bounds for General Linear Operators**”, and this can be helped if L can be written as a difference between two positive linear operators LA and LB , so that $L(f) = LA(f) - LB(f)$.¹⁷ See the “**Example**” section later in this document.

Meanwhile, bounds for the derivatives of f (here, $f^{(i)}$) are often assumed to be known beforehand.

7 Results on Error Bounds

Some results on error bounds for certain classes of operators.

7.1 Bounds for General Positive Linear Operators

The following results give bounds that apply to large classes of positive linear operators. While these operators have relatively simple approximation behavior, in general they do not approximate functions with more than two continuous derivatives “better” than functions with only two.

In this section:

- $\sigma_i = L((e_i - \lambda)^i)(\lambda)$ (the i -th “central moment” of the linear operator L in question).
- $\tau_i = L(\text{abs}(e_i - \lambda)^i)(\lambda)$ (the i -th “absolute moment” of the linear operator L in question).
- $\omega_1(f, \delta)$ is the smallest modulus of continuity of a function f of order 1, with parameter δ .
- $\tilde{\omega}_1(f, \delta)$ is the smallest concave modulus of continuity of f of order 1, both with parameter δ .

Lemma 1. *Let $f(\lambda)$ be continuous on a compact interval, and let L be a positive linear operator that maps continuous functions on that interval to functions of that kind and reproduces all constants (so that $L(e_0) = 1$). Then:*

No.	$\text{abs}(L(f)(\lambda) - f(\lambda)) \leq \dots$
1	$\tilde{\omega}_1(f, \tau_1)$.
2	$2\omega_1(f, (\sigma_2)^{1/2})$.
3	$(1 + (\sigma_2)^{1/2}/h)\omega_1(f, h)$.
4	$(1 + (\sigma_2)/h^2)\omega_1(f, h)$.
5	(Use ineq. 3 if $h < (\sigma_2)^{1/2}$, or ineq. 4 otherwise.)

¹⁶Pițul, P., “Evaluation of the Approximation Order by Positive Linear Operators”, dissertation, Universität Duisberg-Essen, 2007.

¹⁷I suspect that, whenever L is a bounded linear operator that maps continuous functions on a compact interval to functions of that kind, L can be written as a difference between two positive linear operators. But I have not seen a proof of that statement; Acu et al. (“**Grüss-type and Ostrowski-type inequalities in approximation theory**”, Ukr Math J 63, 843–864, 2011) give a similar statement but without proof. <https://doi.org/10.1007/s11253-011-0548-2>

No.	$\text{abs}(L(f)(\lambda) - f(\lambda)) \leq \dots$
6	$\tilde{\omega}_1(f, (\sigma_2)^{1/2})$.

Proof: Inequality 1 follows from a result of Gonska and Meier (1985, theorem 3.1)¹⁸. Inequality 2 follows from a result of Shisha and Mond (1968, theorem 1)¹⁹; inequality 4 comes from another result in the same paper (see also Mamedov (1959)²⁰); inequality 3 follows from a result of Mond (1978)²¹; inequality 5, a result of Păltănea (2004, corollary 1.2.2)²²; inequality 6, a result of Peetre (1969)²³ (also mentioned in Gonska (1998/2023)²⁴, which has an extensive discussion on error bounds for linear operators). []

Remark 1: The moduli of continuity $\omega_1(f, \delta)$ and $\tilde{\omega}_1(f, \delta)$ offer concise ways to express different error bounds depending on how “regular” f is. Properties of these moduli are given in Sevy 1991²⁵, sec. 2.0.2; Gonska 1985²⁶. For example, let f be continuous on a compact interval. Then:

- $\omega_1(f, \delta) \leq \tilde{\omega}_1(f, \delta)$ (Peetre 1969)²⁷.
- If f is Hölder continuous with Hölder exponent α ($0 < \alpha \leq 1$) and Hölder constant M or less, $\omega_1(f, \delta) \leq \tilde{\omega}_1(f, \delta) \leq M\delta^\alpha$. Indeed, in this case, f admits the continuous and concave modulus of continuity $\omega_1(\delta) = M\delta^\alpha$, where $\delta > 0$.
- If f is Lipschitz continuous with Lipschitz constant M or less, $\omega_1(f, \delta) \leq \tilde{\omega}_1(f, \delta) \leq M\delta$, given that Lipschitz-continuous functions are Hölder continuous with Hölder exponent 1. The same bound holds true if f instead has a continuous derivative with maximum absolute value M or less, since in this case (by a result of Hardy and Littlewood) f is Lipschitz continuous with Lipschitz constant M or less.

[]

Example: Let f and L be as in Lemma 1. If f is Lipschitz continuous with Lipschitz constant M or less, or has a continuous derivative with maximum absolute value M or less, $\text{abs}(L(f)(\lambda) - f(\lambda)) \leq M(\sigma_2)^{1/2}$; this follows from the combination of Remark 1 and inequality 6 of Lemma 1.

Lemma 2. *Let $f(\lambda)$ be continuous on a compact interval, and let L be a positive linear operator that maps continuous functions on that interval to functions of that kind and reproduces all polynomials up to degree 1 (constants and linear functions). Let $h > 0$ be a real number. Then:*

No.	If $f \dots$	Then $\text{abs}(L(f)(\cdot) - f(\cdot)) \dots$
1	Has a continuous derivative.	$((h+2)^2/(8h)) \cdot \omega_1(f^{(1)}, h \cdot \sqrt{\sigma_2}) \cdot \sqrt{\sigma_2}$.
2	Has a continuous derivative.	$\frac{1}{2}(\sigma_2)^{1/2} \tilde{\omega}_1(f^{(1)}, (\sigma_2)^{1/2})$.
3	Has a Hölder-continuous derivative with Hölder exponent α ($0 < \alpha \leq 1$) and Hölder constant M or less.	$\frac{M}{2}(\sigma_2)^{(1+\alpha)/2}$.

¹⁸Gonska, H.H., Meier, J., “On approximation by Bernstein-type operators: best constants”, *Studia Sci. Math. Hungar.* 22, 1987.

¹⁹Shisha, O., Mond, B., “The degree of convergence of linear positive operators”, 1968.

²⁰R. G. Mamedov, “On the order of approximation of functions by linear positive operators” (Russian), *Dokl. Akad. Nauk SSSR* 128 (1959).

²¹Mond, B., “On the degree of approximation by linear positive operators”, *Journal of Approximation Theory* 18 (1976).

²²Păltănea, R., *Approximation Theory Using Positive Linear Operators*, Birkhäuser, 2004. <https://doi.org/10.1007/978-1-4612-2058-9>

²³Peetre, J., “On the connection between the theory of interpolation spaces and approximation theory”, in *Approximation Theory*, 1969.

²⁴Gonska, Heiner. “The rate of convergence of bounded linear processes on spaces of continuous functions.” *Journal of Numerical Analysis and Approximation Theory* 52.2 (2023): 182-232. <https://doi.org/10.33993/jnaat522-1326>

²⁵Sevy, J., “Acceleration of convergence of sequences of simultaneous approximants”, dissertation, Drexel University, 1991. <https://doi.org/10.17918/00010296>

²⁶H. H. Gonska, *Quantitative Approximation in $C(X)$* , Habilitationsschrift, Universität Duisburg, 1985.

²⁷Peetre, J., “Exact interpolation theorems for Lipschitz-continuous functions”, *Ricerche Mat.* 18 (1969).

No.	If $f \dots$	Then $\$abs(L(f)(\cdot)-f(\cdot)) \dots \$$
4	Has a Lipschitz-continuous derivative with Lipschitz constant M or less, or has a continuous second derivative with maximum absolute value M or less.	$\frac{M}{2}(\sigma_2)$.

Proof: Inequality 1 is a special case of Theorem 2.19 (in conjunction with Remark 2.21) of Anastassiou (1985), with the interval $[a, b]$, $m = 1$ (since the function is defined on all of $[a, b]$), $r = h$, and x_0 equal to λ . Inequality 2 follows from a result of Gonska and Meier (1985, theorem 4.1)²⁸; see also Păltănea and Dimitriu (2016, remark 3)²⁹. Inequalities 3 and 4 follow from inequality 2 because of Remark 1. []

Lemma 3. *Let $f(\lambda)$ have a continuous k -th derivative on a compact interval, and let L be a positive linear operator that maps continuous functions on that interval to functions of that kind. Let $h > 0$ be a real number. Then $L(f)(\lambda) = L(Q_k(f, \lambda))(\lambda) + L(R_k(f, \lambda))(\lambda)$, where:*

$$abs(L(Q_k(f, \lambda)) - f(\lambda)) \leq \left(\sum_{i=0}^k \frac{\max(abs(f^{(i)}))abs(\sigma_i)}{i!} \right),$$

$$abs(L(R_k(f, \lambda))) \leq \left(\frac{\tau_k}{k!} + \frac{\tau_{k+1}}{(k+1)! \cdot h} \right) \cdot \omega_1(f^{(k)}, h),$$

$$\text{and } abs(L(R_k(f, \lambda))) \leq \max \left(\frac{\tau_k}{k!}, \frac{\tau_{k+1}}{(k+1)! \cdot 2h} \right) \cdot \tilde{\omega}_1(f^{(k)}, 2h),$$

$$\text{and } abs(L(R_k(f, \lambda))) \leq \frac{\tau_k}{k!} \cdot \tilde{\omega}_1(f^{(k)}, \frac{\tau_{k+1}}{(k+1)\tau_k}),$$

and where:

- $Q_k(f, \lambda) = \sum_{i=0}^k f^{(i)}(\lambda) \cdot (e_0 - \lambda)^i / (i!)$ is the degree- k Taylor polynomial of f centered at λ .
- $R_k(f, \lambda)$ is the Taylor remainder that results from subtracting $Q(f, \lambda)$ from f .

Proof: The second to fourth bounds given relate to the Taylor remainder. The second bound comes from Păltănea and Smuc (2019, Theorem 1)³⁰; the third bound comes from corollary 3.2 of Dimitriu (2010)³¹ and Brudnyĭ's lemma; and the fourth bound follows from the second with $h = \tau_{k+1}/(2(k+1)\tau_k)$ and comes from Gonska et al. (2006)³², where the compact interval assumed was the closed unit interval; see also Gonska

²⁸Gonska, H.H., Meier, J., "On approximation by Bernstein-type operators: best constants", *Studia Sci. Math. Hungar.* 22, 1987.

²⁹Păltănea, R., Dimitriu, M.T., "On some second order moduli of smoothness." *General Mathematics* 24 (2016)

³⁰Păltănea, R., Smuc, M. "Sharp Estimates of Asymptotic Error of Approximation by General Positive Linear Operators in Terms of the First and the Second Moduli of Continuity", *Results in Mathematics* 74, 70 (2019). <https://doi.org/10.1007/s00025-019-0997-8>

³¹Dimitriu, M.T., "Estimates with optimal constants using Peetre's K-functionals", *Carpathian Journal of Mathematics* 26 (2010). <https://www.jstor.org/stable/43964559>

³²Gonska, Heiner, Paula Pițul, and Ioan Rașu. "On Peano's form of the Taylor remainder, Voronovskaja's theorem and the commutator of positive linear operators". In *Proceedings of the International Conference on Numerical Analysis and Approximation Theory*, Cluj-Napoca. Romania, July 2006.

(2007)³³, Pițul (2007)³⁴. See also Anastassiou (1985, theorem 2.31)^{35,36} []

Lemma 4. *Let k be zero or a positive integer. Let $f(\lambda)$ —*

1. *have a Lipschitz-continuous k -th derivative on a compact interval, with Lipschitz constant M or less, or*
2. *have a continuous $(k+1)$ -th derivative on that interval, with maximum absolute value M or less,*

and let L be a positive linear operator that maps continuous functions on that interval to functions of that kind. Then $\text{abs}(L(R_k(f, \lambda))) \leq M\tau_{k+1}/((k+1)!)$, where $R_k(f, \lambda)$ is as in Lemma 3.

Proof: Follows from the third bound for $L(R_k(f, \lambda))$ in Lemma 3 in the same manner as inequality 10 of Lemma 2, using Remark 1. []

The following two lemmas are more general, but not as easy to use.

Lemma 4A (special case of Theorem 3.4 in Gonska (1998/2023)³⁷). *Let $f(\lambda)$ be continuous on a compact interval or a closed subset thereof, and let L be a positive linear operator that maps continuous functions on f 's domain to bounded functions on that domain. Let $h > 0$ be a real number. Then:*

$$\begin{aligned} \text{abs}(L(f)(\lambda) - f(\lambda)) &\leq \max(\|L(e_0)\|, L(\text{abs}(e_1 - \lambda))(\lambda)) \cdot \tilde{\omega}_1(f, h) \\ &\quad + \text{abs}(L(e_0)(\lambda) - 1) \cdot \text{abs}(f(\lambda)) \\ &\leq (\|L(e_0)\| + L(\text{abs}(e_1 - \lambda))(\lambda)) \cdot \tilde{\omega}_1(f, h) + \text{abs}(L(e_0)(\lambda) - 1) \cdot \text{abs}(f(\lambda)), \end{aligned}$$

where $\|L(e_0)\|$ is the maximum of $L(e_0)$ over f 's domain.

Lemma 4B (special case of Theorem 4.7 in Gonska (1998/2023)³⁸). *Let $f(\lambda)$ be continuous on a compact interval, and let L be a positive linear operator that maps bounded functions on f 's domain to bounded functions on that domain. Let $h > 0$ be a real number. Then:*

$$\begin{aligned} \text{abs}(L(f)(\lambda) - f(\lambda)) &\leq (L(e_0)(\lambda) + L(\text{ceil}((e_1 - \lambda)/h - 1))(\lambda)) \cdot \omega_1(f, h) \\ &\quad + \text{abs}(L(e_0)(\lambda) - 1) \cdot \text{abs}(f(\lambda)), \\ \text{abs}(L(f)(\lambda) - f(\lambda)) &\leq (L(e_0)(\lambda) + L(\text{abs}(e_1 - \lambda))(\lambda)/h) \cdot \omega_1(f, h) \end{aligned}$$

³³Gonska, Heiner. “On the degree of approximation in Voronovskaja’s theorem”, *Studia Univ. Babeş-Bolyai, Math.*, September 2007.

³⁴Pițul, P., “Evaluation of the Approximation Order by Positive Linear Operators”, dissertation, Universität Duisberg-Essen, 2007.

³⁵Anastassiou, George A. “A study of positive linear operators by the method of moments, one-dimensional case.” *Journal of Approximation Theory* 45.3 (1985): 247-270. <https://www.sciencedirect.com/science/article/pii/0021904585900498>

³⁶The paper Cichoń et al., “On delta-method of moments and probabilistic sums”, *ANALCO* 2013, has very similar results, but they assume the function f has a k -th derivative defined on an *open* interval (say, $0 < \lambda < 1$), rather than a *compact* interval, making those results harder to use if Y is a random variable that can take a value equal to either endpoint of the interval (in this example, 0 or 1). <https://doi.org/10.1137/1.9781611973037.11>

³⁷Gonska, Heiner. “The rate of convergence of bounded linear processes on spaces of continuous functions.” *Journal of Numerical Analysis and Approximation Theory* 52.2 (2023): 182-232. <https://doi.org/10.33993/jnaat522-1326>

³⁸Gonska, Heiner. “The rate of convergence of bounded linear processes on spaces of continuous functions.” *Journal of Numerical Analysis and Approximation Theory* 52.2 (2023): 182-232. <https://doi.org/10.33993/jnaat522-1326>

$$+\text{abs}(L(e_0)(\lambda) - 1) \cdot \text{abs}(f(\lambda)).$$

The second inequality also works if L maps from continuous functions instead of from bounded functions.

Notes:

1. Using Lemma 4A requires calculating $L(e_0)$, $\|L(e_0)\|$, and $L(\text{abs}(e_1 - \lambda))$, or finding upper bounds for these.
2. Using Lemma 4B requires calculating $L(e_0)$ and either $L(\text{abs}(e_1 - \lambda))$ or $L(\text{ceil}((e_1 - \lambda)/h - 1))$, or finding upper bounds for these values.
3. Unlike Lemma 4A, Lemma 4B is not guaranteed to work if f 's domain is a closed subset of an interval (see Remark 2.5 in Gonska (1998/2023)³⁹).

The following lemma adapts the previous lemmas to the setting of random variables.

Lemma 5. *Let $f(\lambda)$ be continuous on a compact interval, and let Y be a random variable taking only values in that interval. Then Lemmas 1 through 4A apply as appropriate to f meeting their conditions, with $L(f) = \mathbb{E}[f(Y)]$ and $\lambda = \mathbb{E}[Y]$.*

Proof: With these assumptions there is a positive linear operator $L(f) = \mathbb{E}[f(Y)]$ for Y and f , according to Theorem 3.1.1 of Frantz (1984)⁴⁰, letting $x_o = \lambda$. Then $L(e_0) = \mathbb{E}[e_0(Y)] = \mathbb{E}[1] = 1$ regardless of Y , and $L(e_1) = \mathbb{E}[e_1(Y)] = \mathbb{E}[Y] = \lambda$, so L reproduces all polynomials of degree up to 1. []

7.2 Bounds for Remainder of Bernstein Polynomials

The following results specialize the previous ones to the case of **Bernstein polynomials** B_n . They apply to the Bernstein polynomial of the result of subtracting a Taylor polynomial from a function, and are useful when a linear operator contains $B_n(f)$ in its definition and reproduces all polynomials of degree r or less.

Lemma 6: *Let k be zero or a positive integer. Let $f(\lambda)$ —*

1. *have a Lipschitz-continuous k -th derivative on the closed unit interval, with Lipschitz constant M or less, or*
2. *have a continuous $(k + 1)$ -th derivative on that interval, with maximum absolute value M or less.*

Then the following bound holds true: $\text{abs}(B_n(R_k(f, \lambda))) \leq (M\mu_{k+1})/(((k + 1)!)n^{(k+1)/2})$ for every integer $n \geq 2$ (and also for $n = 1$ if k is odd), where μ_k is as defined in Proposition 1.

Proof: Follows from Lemma 4, with $L(f) = B_n(f)$, and from Proposition 1. []

Corollary 1: *Let $f(\lambda)$, k , and M be as in Lemma 6. Then, for every $0 \leq \lambda \leq 1$:*

If k is:	Then $\text{abs}(B_n(R_k(f, \lambda))) \leq \dots$
0.	$M(1/2)/n^{1/2}$ for every integer $n \geq 1$.
1.	$M(1/8)/n = 0.125M/n$ for every integer $n \geq 1$.
2.	$M(\sqrt{3}/48)/n^{3/2} < 0.3609M/n^{3/2}$ for every integer $n \geq 2$.
3.	$M(1/128)/n^2 = 0.0078125M/n^2$ for every integer $n \geq 1$.
4.	$M(\sqrt{5}/1280)/n^{5/2} < 0.001747/n^{5/2}$ for every integer $n \geq 2$.
5.	$M(1/3072)/n^3 < 0.0003256/n^3$ for every integer $n \geq 1$.

³⁹Gonska, Heiner. “The rate of convergence of bounded linear processes on spaces of continuous functions.” Journal of Numerical Analysis and Approximation Theory 52.2 (2023): 182-232. <https://doi.org/10.33993/jnaat522-1326>

⁴⁰Frantz, Deborah A. **Summability methods, probability distributions, and associated positive linear operators.** Lehigh University, 1984. <https://preserve.lehigh.edu/lehigh-scholarship/graduate-publications-theses-dissertations/theses-dissertations/summability>

7.3 Bounds for General Linear Operators

The results in this section give error bounds for important classes of linear operators (not necessarily positive ones). But, in general, they are harder to use than the ones for positive linear operators, because more has to be computed for nonpositive operators than just the “moments”.

Roughly speaking, the *integral* of $f(\lambda)$ on the compact interval $[a, b]$ is the “area under the graph” of that function when the function is restricted to that interval. If f is continuous there, this is the value that—

$$\frac{b-a}{n} \sum_{i=1}^n f\left(a + (b-a)\left(i - \frac{1}{2}\right)/n\right), \quad (2A)$$

approaches as n gets larger and larger. The integral of $f(\lambda)$ on $[a, b]$ is denoted $\int_a^b f(\lambda) d\lambda$.

The next two lemmas rely on the so-called *Peano kernel theorem*, which was originally developed to assess the error in estimating the integral of a function from samples of it⁴¹ (for more on this theory, see Brass and Förster 1998⁴²; Waldron 1999⁴³).

Lemma 7. *Let k be zero or a positive integer, let $f(\lambda)$ have a continuous $(k+1)$ -th derivative on the compact interval $[a, b]$. Let C and c be real numbers such that $c \leq f^{(k+1)} \leq C$ over that interval. Let L be a bounded linear operator that—*

- *reproduces all polynomials of degree k or less, and*
- *maps continuous functions (or, if $k = 0$, bounded functions) on the interval $[a, b]$ to continuous functions on that interval.*

Then:

$$\text{abs}(LF(f)(\lambda)) = \text{abs}(f(\lambda) - L(f)(\lambda))$$

$$\leq \frac{C-c}{2} \int_a^b \text{abs}(LF((e_1 - t)_+^k)(\lambda)/(k!)) dt \quad (3)$$

$$= \frac{C-c}{2(k!)} \int_a^b \text{abs}(LF((e_1 - t)_+^k)(\lambda)) dt \quad (4)$$

$$\begin{aligned} &= \frac{C-c}{2(k!)} \left(\int_a^\lambda \text{abs}(LF((e_1 - t)_+^k)(\lambda)) dt + \int_\lambda^b \text{abs}(L((e_1 - t)_+^k)(\lambda)) dt \right) \\ &= \frac{C-c}{2(k!)} \left(\int_a^\lambda \text{abs}((\lambda - t)^k - L((e_1 - t)_+^k)(\lambda)) dt + \int_\lambda^b \text{abs}(L((e_1 - t)_+^k)(\lambda)) dt \right), \\ &\text{abs}(LF(f)(\lambda)) \leq \frac{\|f^{(k+1)}\|}{k!} \int_a^b \text{abs}(LF((e_1 - t)_+^k)(\lambda)) dt, \end{aligned} \quad (5)$$

⁴¹This kind of estimation is called *quadrature* or *numerical integration*, and methods for such estimation, such as the one that is given in (2A) and also known as the midpoint rule, are called *quadrature rules* (see also Brass, H., Petras, K., *Quadrature Theory: The Theory of Numerical Integration on a Compact Interval*, American Mathematical Society, 2011).

⁴²Brass, H., Förster, K.J. (1998). On the Application of the Peano Representation of Linear Functionals in Numerical Analysis. In: Milovanović, G.V. (eds) Recent Progress in Inequalities. Mathematics and Its Applications, vol 430. Springer, Dordrecht. https://doi.org/10.1007/978-94-015-9086-0_10

⁴³Waldron, Shayne. “Refinements of the Peano kernel theorem.” Numerical functional analysis and optimization 20.1-2 (1999): 147-161. <https://doi.org/10.1080/01630569908816885>

where $LF(f) = f - L(f)$, and the notation $(x)_+^k$ is as follows. If $k > 0$, this equals $((x + \text{abs}(x))/2)^k$, or $\max(0, x)^k$, and if k is 0, this equals either 1 if $x \geq 0$ or 0 otherwise.

Formulas (3) and (4) are because, in this case, the operator LF equals 0 for every polynomial of degree k or less, so that $LF(e_i) = 0$ whenever $0 \leq i \leq k$, so that LF satisfies theorem 3 of Gavrea and Ivan (2015)⁴⁴. Formula (5) is an easy consequence of (4); see also Brass and Förster (1998, theorem 5)^{45, 46}.

Note: The operator—

$$\frac{LF((e_1 - t)_+^k)}{k!} = \frac{(e_1 - t)_+^k - L((e_1 - t)_+^k)}{k!},$$

where $k \geq 0$, is called the *Peano kernel of order $k + 1$* of LF (Brass and Förster 1998)⁴⁷, with a fixed value of t such that $a \leq t \leq b$. But finding a “closed form” of Peano kernels is relatively hard compared to “raw moments” and “central moments”. Luckily, only an upper bound of the Peano kernel is needed to use the formulas in this lemma.

Lemma 8 (see Theorem 4 of Gavrea and Ivan (2015)⁴⁸). *With the assumptions in Lemma 7, if LF is the difference of two positive linear operators LA and LB , so that $LF(f) = LA(f) - LB(f)$ (or $L(f) = f - LA(f) + LB(f)$), and LA and LB both map continuous functions on that interval to functions of that kind, then:*

$$\text{abs}(L(f)(\lambda) - f(\lambda)) \leq \frac{C - c}{(k + 1)!} \text{abs}(LA(e_{k+1})(\lambda)),$$

$$\text{abs}(L(f)(\lambda) - f(\lambda)) \leq \frac{2\|f^{(k+1)}\|}{(k + 1)!} \text{abs}(LA(e_{k+1})(\lambda)).$$

Lemma 9 (special case of Theorem 3.2 in Gonska (1998/2023)⁴⁹). *Let $f(\lambda)$ be continuous on a compact interval or a closed subset thereof, and let L be a bounded linear operator that maps continuous functions on f 's domain to bounded functions on that domain. Let $h > 0$ be a real number. Then for each λ in f 's domain:*

$$\text{abs}(L(f)(\lambda) - f(\lambda)) \leq \max((\|L\| + \alpha)/2, (\gamma(\beta(\lambda) - L(e_0)(\lambda)) + \text{abs}(L(\text{abs}(e_1 - \lambda))(\lambda))))/h$$

$$\cdot \tilde{\omega}_1(f, h) + \text{abs}(L(e_0)(\lambda) - 1) \cdot \text{abs}(f(\lambda)),$$

⁴⁴Gavrea, I., Ivan, M., “A sharp estimate for the Peano error representation”, *Applied Mathematics and Computation* 252 (2015). <https://doi.org/10.1016/j.amc.2014.12.017>

⁴⁵Brass, H., Förster, K.J. (1998). On the Application of the Peano Representation of Linear Functionals in Numerical Analysis. In: Milovanović, G.V. (eds) *Recent Progress in Inequalities. Mathematics and Its Applications*, vol 430. Springer, Dordrecht. https://doi.org/10.1007/978-94-015-9086-0_10

⁴⁶Note that for formulas (3) to (5), $(e_1 - t)_+^0$ is discontinuous and so is not accepted by LF and L if they map from only continuous functions; thus the results in this section suppose both operators map from bounded functions for $k = 0$. Brass and Förster 1998 adequately provides for the case $k = 0$, but not Gavrea and Ivan 2015, unfortunately.

⁴⁷Brass, H., Förster, K.J. (1998). On the Application of the Peano Representation of Linear Functionals in Numerical Analysis. In: Milovanović, G.V. (eds) *Recent Progress in Inequalities. Mathematics and Its Applications*, vol 430. Springer, Dordrecht. https://doi.org/10.1007/978-94-015-9086-0_10

⁴⁸Gavrea, I., Ivan, M., “A sharp estimate for the Peano error representation”, *Applied Mathematics and Computation* 252 (2015). <https://doi.org/10.1016/j.amc.2014.12.017>

⁴⁹Gonska, Heiner. “The rate of convergence of bounded linear processes on spaces of continuous functions.” *Journal of Numerical Analysis and Approximation Theory* 52.2 (2023): 182-232. <https://doi.org/10.33993/jnaat522-1326>

where $\|L\|$ is the operator norm of L , α is the maximum of $\text{abs}(L(e_0))$ over f 's domain; $\beta(\lambda)$ is the maximum of $\text{abs}(L(g)(\lambda))$ over all continuous functions g on f 's domain with a maximum absolute value of 1 or less; and γ is the difference between the highest and lowest value of λ in f 's domain.

Lemma 10. *With the assumptions in Lemma 9, if L reproduces constants, so that $L(e_0) = 1$, the inequality in that lemma becomes:*

$$\text{abs}(L(f)(\lambda) - f(\lambda)) \leq \max((1 + \|L\|)/2, (\gamma(\beta(\lambda) - 1) + \text{abs}(L(\text{abs}(e_1 - \lambda))(\lambda)))/h) \cdot \tilde{\omega}_1(f, h).$$

Lemma 11 (special case of Theorem 4.4 and Corollary 4.5 in Gonska (1998/2023)⁵⁰). *Let $f(\lambda)$ be continuous on a compact interval $[a, b]$, and let L be a bounded linear operator that maps continuous functions on f 's domain to bounded functions on that domain. Let $h > 0$ be a real number. Then for each λ in f 's domain:*

$$\begin{aligned} \text{abs}(L(f)(\lambda) - f(\lambda)) &\leq ((\beta(\lambda) - \text{abs}(L(e_0)(\lambda))) \cdot (1 + (b - a)/h) \\ &\quad + \text{abs}(L(e_0)(\lambda)) + \text{abs}(L(\text{abs}(e_1 - \lambda))(\lambda))/h) \cdot \omega_1(f, h) \\ &\quad + \text{abs}(L(e_0)(\lambda) - 1) \cdot \text{abs}(f(\lambda)), \end{aligned}$$

where $\beta(\lambda)$ is as in Lemma 9.

Lemma 12. *With the assumptions in Lemma 11, if L reproduces constants, so that $L(e_0) = 1$, the inequality in that lemma becomes:*

$$\begin{aligned} \text{abs}(L(f)(\lambda) - f(\lambda)) &\leq ((\beta(\lambda) - 1) \cdot (1 + (b - a)/h) \\ &\quad + 1 + \text{abs}(L(\text{abs}(e_1 - \lambda))(\lambda))/h) \cdot \omega_1(f, h) \end{aligned}$$

7.4 Whitney's Inequality on Polynomial Errors

The following inequality gives a bound on the “best possible” error that a polynomial of degree n can achieve in approximating a function.

Let n be zero or a positive integer, let $f(\lambda)$ be continuous on a compact interval $[a, b]$, and let P be a polynomial of degree n or less with the least maximum absolute difference between f and the polynomial on that interval. Then the error of P in approximating f is bounded as follows (see Babenko and Kryakin 2019⁵¹):

$$\|f - P\| \leq W \cdot \omega_{n+1}(f, \frac{b-a}{n+1}),$$

where—

- W is:
 - 1 if $n \leq 7$.

⁵⁰Gonska, Heiner. “The rate of convergence of bounded linear processes on spaces of continuous functions.” *Journal of Numerical Analysis and Approximation Theory* 52.2 (2023): 182-232. <https://doi.org/10.33993/jnaat522-1326>

⁵¹Babenko, Alexander G., and Yuriy V. Kryakin. “Special difference operators and the constants in the classical Jackson-type theorems.” *Topics in Classical and Modern Analysis: In Memory of Yingkang Hu*. Cham: Springer International Publishing, 2019. 35-46.

- $(2 + \exp(-2))(< 2.13534)$ if $n \geq 8$.
- $3/4$ if $n = 1$ and f is convex (Singh Kaire and Prymak 2023/2025)⁵².
- $1/2$ if $n = 1$, f is convex, and $a = -b$ (Singh Kaire and Prymak 2023/2025)⁵³.
- $\omega_n(f, h)$ is the smallest modulus of continuity of f of order n , with parameter h .

Using properties of moduli of continuity (see Sevy 1991⁵⁴, sec. 2.0.2; Gonska 1985⁵⁵), if f has a continuous $(n + 1)$ -th derivative on $[a, b]$:

$$\|f - P\| \leq W \cdot \left(\frac{b-a}{n+1}\right)^{n+1} \|f^{(n+1)}\|,$$

and if f has a continuous n -th derivative on that interval:

$$\|f - P\| \leq W \cdot \left(\frac{b-a}{n+1}\right)^n \omega_1(f^{(n)}, \frac{b-a}{n+1}).$$

7.5 Another Inequality on Polynomial Errors

Like Whitney’s inequality, the following gives a bound on the “best possible” error between a polynomial and a function.

Let n be zero or a positive integer, let $f(\lambda)$ have a continuous $(n + 1)$ -th derivative on the compact interval $[a, b]$, and let P be a polynomial of degree n or less with the least maximum absolute difference between f and the polynomial on that interval. Then the error of P in approximating f is bounded as follows (Brass and Petras 2011⁵⁶, example 3.1.2; Phillips 2003⁵⁷, theorem 2.4.6):

$$\|f - P\| \leq \frac{(b-a)^r}{((n+1)!) \cdot 2^{2n+1}} \|f^{(n+1)}\|.$$

7.6 Lebesgue Inequality for Certain Linear Operators

Let $f(\lambda)$ be a continuous function on a compact interval. Let L be a linear operator that—

1. maps continuous or bounded functions on that interval to functions in a finite-dimensional subspace of continuous or bounded functions (such as polynomials up to degree 5)⁵⁸, and
2. is *idempotent*, that is, applying the linear operator twice or more is the same as applying it once, so that $L(L(f)) = L(f)$ for every allowed function f .⁵⁹

⁵²Jaskaran Singh Kaire and Andriy Prymak, “**Whitney-type estimates for convex functions**”, arXiv:2311.00912 (2023). <https://arxiv.org/abs/2311.00912>

⁵³Jaskaran Singh Kaire and Andriy Prymak, “**Whitney-type estimates for convex functions**”, arXiv:2311.00912 (2023). <https://arxiv.org/abs/2311.00912>

⁵⁴Sevy, J., “Acceleration of convergence of sequences of simultaneous approximants”, dissertation, Drexel University, 1991. <https://doi.org/10.17918/00010296>

⁵⁵H. H. Gonska, *Quantitative Approximation in $C(X)$* , Habilitationsschrift, Universität Duisburg, 1985.

⁵⁶Brass, H., Petras, K., *Quadrature Theory: The Theory of Numerical Integration on a Compact Interval*, American Mathematical Society, 2011

⁵⁷Phillips, G.M., *Interpolation and Approximation by Polynomials*, Springer New York, NY, 2003. <https://doi.org/10.1007/b97417>

⁵⁸A subset of functions is a *subspace* if it has the following property: if functions f and g are in the family, so are $f + g$, $f + \alpha$, and $g + \alpha$ for every real number α (Weisstein, Eric W. “Subspace.” From MathWorld—A Wolfram Resource. <https://mathworld.wolfram.com/Subspace.html>). The subspace is also *finite-dimensional* if it is the smallest subspace that contains a given finite set of functions. One example of a finite-dimensional subspace of continuous functions is the polynomials up to degree 2, which are the smallest subspace containing e_0 , e_1 , and e_2 .

⁵⁹This includes the case that L reproduces all functions it maps to (for example, if L maps to polynomials up to degree 5, it reproduces all such polynomials).

Also, let I be the identity operator $I(f) = f$.

Then the following error bounds (also known as *Lebesgue's lemma* or the *Lebesgue inequality*) hold true for every function P mapped to by L :

$$\text{abs}(f(x) - L(f)(x)) \leq (1 + \|L\|) \cdot \max_t(\text{abs}(f(t) - P(t))), \quad (\text{Leb})$$

$$\text{abs}(f(x) - L(f)(x)) \leq \|I - L\| \cdot \max_t(\text{abs}(f(t) - P(t))), \quad (\text{Leb2})$$

where $\|L\|$ is the operator norm of L , and $\|I - L\|$ is the operator norm for the difference between f and $L(f)$ (see also DeVore and Lorentz (1993, p. 30; ch. 5)⁶⁰, Cheney (1996, chapter 6)⁶¹, Powell (1981, theorem 3.1)⁶², De Villiers (2012, theorem 5.3.2)⁶³; for the second inequality see De Boor (1982, chapter 2)⁶⁴).

If every function mapped to by L is a polynomial, though, this error bound will generally be crude or trivial unless L_n are nonpositive operators. Indeed, the only positive linear operator L that reproduces all polynomials up to degree 2 is the identity operator.⁶⁵

Examples:

1. Let f have a continuous third derivative on the closed unit interval. Combining the inequality (Leb) with the Whitney-type inequalities given earlier leads to the following error bound for idempotent linear operators L that map continuous functions on that interval to polynomials up to degree 2:

$$\begin{aligned} \text{abs}(L(f)(x) - f(x)) &\leq (1 + \|L\|) \cdot 1 \cdot \left(\frac{1}{3}\right)^3 \|f^{(3)}\| \\ &= (1 + \|L\|) \|f^{(3)}\| / 27. \end{aligned}$$

2. The Bernstein polynomial B_n maps continuous functions to polynomials up to degree n , but if n is 2 or greater it is *not* idempotent because it does not reproduce all polynomials up to degree n . Thus, the inequalities (Leb) and (Leb2) do not apply to Bernstein polynomials of degree 2 or greater.
3. The *identity operator* $I(f) = f$ has operator norm 1 and maps every function to itself. For this operator, the inequality (Leb2) becomes:

$$\text{abs}(I(f)(x) - f(x)) \leq \|I - I\| \cdot 0 = 0.$$

Note: Inequalities similar to (Leb) and (Leb2) may apply to *spline operators* that map to a continuous function (a *spline*) that equals a polynomial at subintervals of its domain (for example,

⁶⁰R.A. DeVore and G.G. Lorentz, *Constructive Approximation*, 1993. <https://link.springer.com/book/9783540506270>

⁶¹E. W. Cheney, *Introduction to Approximation Theory*, 1998.

⁶²Powell, Michael James David. *Approximation theory and methods*. Cambridge University Press, 1981.

⁶³De Villiers, J., *Mathematics of Approximation*, Atlantis Press, 2012.

⁶⁴De Boor, C. (1982). Topics in multivariate approximation theory. In: Turner, P.R. (eds) *Topics in Numerical Analysis*. Lecture Notes in Mathematics, vol 965. Springer, Berlin, Heidelberg. <https://doi.org/10.1007/BFb0063200>

⁶⁵Guessab, A., Nouisser, O. & Schmeisser, G. Enhancement of the algebraic precision of a linear operator and consequences under positivity. *Positivity* 13, 693–707 (2009). <https://doi.org/10.1007/s11117-008-2253-4>. However, Gavrea and Ivan (“**A note on the fixed points of positive linear operators**”, *Journal of Approximation Theory* (227), 2018) pointed out that there are positive linear operators besides the identity that reproduce all polynomials of the form x^i where $i > 0$. <https://doi.org/10.1016/j.jat.2017.12.001>

Sablonnière 2007⁶⁶). These operators may have a *local approximation* property, such that $L(f)(\lambda)$ depends only on the behavior of f near λ . But it's not clear to me when inequalities similar to (Leb) and (Leb2) apply to those cases.⁶⁷

7.7 Bounds for Certain Nonlinear Operators

The following comes from a result in Bede and Gal (2010)⁶⁸; see also Bede et al. (2009)⁶⁹.

Let $f(\lambda)$ be continuous, bounded, and nonnegative on an interval. Let L be an operator that maps functions of that kind to functions of that kind and also has the following properties:

1. (Monotone.) For every pair of allowed functions g and h , if $g \leq h$, then $L(g) \leq L(h)$.
2. (Subadditive.) For every pair of allowed functions g and h , $L(g + h) \leq L(g) + L(h)$.
3. (Positively homogeneous.) $xL(g) = L(xg)$ for every allowed function g and every $x \geq 0$.

If $L(e_0) = 1$, then for every $h > 0$:

$$\text{abs}(f(x) - L(f)(x)) \leq (1 + L(\text{abs}(e_0 - x))(x)/h) \cdot \omega_1(f, h),$$

provided $L(\text{abs}(e_0 - x))(x)$ (the “absolute moment” of L) exists (and is finite or infinite).

Notes: An operator meeting conditions 2 and 3 is also called a *sublinear* operator. Every linear operator is also sublinear. A linear operator is monotone if and only if it is positive. For more on nonlinear operators, see Gal and Niculescu (2023)⁷⁰; on nonlinear approximation, see DeVore (1998)⁷¹.

8 Example

This example shows how to find a linear operator's bounds.

Let $L_n(f)$ be a linear operator inspired by **a conjecture I have**⁷² on polynomial approximation. It is described as follows:

$$L_n(f)(\lambda) = \sum_{i=0}^n \left(W_{2n}(f) \left(\frac{k}{2n} \right) - W_n(f) \left(\frac{i}{n} \right) \right) \sigma_{n,k,i}$$

⁶⁶Sablonnière, Paul. “A quadrature formula associated with a univariate spline quasi interpolant.” BIT Numerical Mathematics 47.4 (2007): 825-837. <https://doi.org/10.1007/s10543-007-0146-8>

⁶⁷It has been argued that the inequalities in this section apply to spline operators that reproduce every polynomial up to degree d , say, on *subintervals* of their domain even though they map to more functions than polynomials on the *whole* domain. For example, compare—Sablonnière, P. “Univariate spline quasi-interpolants and applications to numerical analysis.” Rend. Sem. Mat. Univ. Pol. Torino 63.3 (2005), section 2, and Sablonnière, P. “Quadratic spline quasi-interpolants on bounded domains of $\mathbb{R}^d$, $d = 1, 2, 3$.” Rend. Sem. Mat. Univ. Pol. Torino 61.3 (2003), matter after Remark 1, with the operators in Lee, B-G., et al., “Some examples of quasi-interpolants constructed from local spline projectors”, Mathematical methods for curves and surfaces, Oslo 2000 (2000), section 2, which were shown to reproduce all functions they map to. But what stops us from subdividing the intervals further into, say, $10n$ subintervals of length $1/(10n)$, and inferring a much smaller error bound? Indeed, in this example, $L(f)$ then still equals a degree-2-or less polynomial at each of the new subintervals and reproduces polynomials of that kind at each of them.

⁶⁸Bede, Barnabás, and Sorin G. Gal. “Approximation by Nonlinear Bernstein and Favard-Szász-Mirakjan Operators of Max-Product Kind.” Journal of Concrete & Applicable Mathematics 8.1 (2010).

⁶⁹Bede, Barnabás, Coroianu, Lucian, Gal, Sorin G., Approximation and Shape Preserving Properties of the Bernstein Operator of Max-Product Kind, International Journal of Mathematics and Mathematical Sciences, 2009, 590589, 26 pages, 2009. <https://doi.org/10.1155/2009/590589>

⁷⁰Gal, Sorin G., and Constantin P. Niculescu. “Korovkin-type theorems for weakly nonlinear and monotone operators”, arXiv:2206.14102v1 [math.FA], also in *Mediterranean Journal of Mathematics* 20.2 (2023): 56. <https://doi.org/10.1007/s00009-023-02271-y> <https://arxiv.org/abs/2206.14102v1>

⁷¹DeVore, R. A. (1998). Nonlinear approximation. Acta Numerica, 7, 51–150. <https://doi.org/10.1017/S0962492900002816>.

⁷²https://peteroupc.github.io/bernsupp.html#A_Conjecture_on_Polynomial_Approximation

$$= \mathbb{E} \left[W_{2n}(f) \left(\frac{k}{2n} \right) - W_n(f) \left(\frac{X_k}{n} \right) \right],$$

where:

- $k = 2n\lambda$, where $0 \leq \lambda \leq 1$.
- $W_n(f)$ is a linear operator that approaches f as n increases.⁷³
- X_k is a hypergeometric($2n, k, n$) random variable.
- $\sigma_{n,k,i}$ equals $\binom{n}{i} \binom{n}{k-i} / \binom{2n}{k}$ and is the probability that X_k equals i .

L_n and W_n are generally nonpositive operators. As an example, take $W_n = 2f - B_n(f)$. Then $B_n(W_n(f))$ is a linear operator that is the iterated Boolean sum of degree- n Bernstein polynomials, with one iteration; see Güntürk and Li (2021a, Theorem 5)⁷⁴. That paper, among others (for example, Micchelli 1973⁷⁵), showed that this operator approaches f at the rate $O(1/n^{3/2})$ if f has a continuous third derivative. (“ $O(1/n^{3/2})$ ” means the error is no greater than a constant times $1/n^{3/2}$ for all values of n .)

With this choice of W_n , L_n becomes:

$$\begin{aligned} L_n(f)(\lambda) &= \sum_{i=0}^n \left(\left(2f \left(\frac{k}{2n} \right) - B_{2n}(f) \left(\frac{k}{2n} \right) \right) - \left(2f \left(\frac{i}{n} \right) - B_n(f) \left(\frac{i}{n} \right) \right) \right) \sigma_{n,k,i} \\ &= \mathbb{E} \left[\left(2f \left(\frac{k}{2n} \right) - B_{2n}(f) \left(\frac{k}{2n} \right) \right) - \left(2f \left(\frac{X_k}{n} \right) - B_n(f) \left(\frac{X_k}{n} \right) \right) \right] \\ &= \sum_{i=0}^n \left(\left(2f \left(\frac{k}{2n} \right) + B_n(f) \left(\frac{i}{n} \right) \right) \right) \sigma_{n,k,i} - \sum_{i=0}^n \left(\left(2f \left(\frac{i}{n} \right) + B_{2n}(f) \left(\frac{k}{n} \right) \right) \right) \sigma_{n,k,i} \\ &= LA_n(f)(\lambda) - LB_n(f)(\lambda). \end{aligned}$$

Here, LA_n and LB_n are positive linear operators, making it easier to assess their approximation properties.

It will be shown that, if f has a continuous third derivative, the rate of L_n towards zero is $O(M/n^{3/2})$, where M is the maximum absolute value of f and its derivatives up to the third derivative. The proof of this relies on exact expressions of L_n ’s “**raw moments**” and “**central moments**”, and those for the combined operator $(LA_n + LB_n)$.

The following are some of these values and those for related operators:

- $L_n(e_0)(x) = L_n((e_1 - x)^0)(x) = 0$.
- $L_n(e_1)(x) = L_n((e_1 - x)^1)(x) = 0$.
- $L_n(e_2)(x) = L_n((e_1 - x)^2)(x) = 3x(x-1)/(2n(2n-1)) = O(1/n^2)$.
- $L_n(e_3)(x) = n^3 x^2(2nx - 4x + 3)/(2n-1)$.
- $LA_n((e_1 - x)^2)(x) = -x(3n-2) \cdot (x-1)/(n(2n-1)) = O(1/n)$.
- $LB_n((e_1 - x)^2)(x) = -x(6n-1) \cdot (x-1)/(2n(2n-1)) = O(1/n)$.
- $(LA_n + LB_n)((e_1 - x)^2)(x) = LA_n(\text{abs}(e_1 - x)^2)(x) + LB_n(\text{abs}(e_1 - x)^2)(x) = -x(12n-5) \cdot (x-1)/(2n(2n-1)) = O(1/n)$.

⁷³ W_n can, in principle, be nonlinear instead, but this would require a totally different approach to finding the approximation error, and L_n would then be nonlinear in general.

⁷⁴Güntürk, C. Sinan, and Weilin Li. “**Approximation with one-bit polynomials in Bernstein form**”, arXiv:2112.09183 (2021); Constr Approx 57, 601–630 (2023). <https://doi.org/10.1007/s00365-022-09608-y> <https://arxiv.org/pdf/2112.09183>

⁷⁵Micchelli, Charles. “**The saturation class and iterates of the Bernstein polynomials**”, Journal of Approximation Theory 8, no. 1 (1973): 1–18. <https://www.sciencedirect.com/science/article/pii/0021904573900282>

To find values like those just listed, it is useful to calculate raw moments (Wang et al. 2023)⁷⁶ and central moments (Weisstein)⁷⁷ of hypergeometric random variables (such as X_k). Indeed, if $g(y) = W_{2n}(e_r; k/(2n)) - W_n(e_r; y)$ is a polynomial in y of degree r or less, then $L_n(e_r)$ can be found using a Taylor expansion, namely as—

$$\begin{aligned} L_n(e_r) &= \sum_{i=0}^r \mathbb{E}[(X_k/n - \mathbb{E}[X_k/n])^i] \frac{g^{(i)}(\mathbb{E}[X_k/n])}{i!} \\ &= \sum_{i=0}^r \frac{\mathbb{E}[(X_k - \mathbb{E}[X_k])^i]}{n^i} \frac{g^{(i)}(k/(2n))}{i!}, \end{aligned}$$

where the derivatives are taken with respect to y , and where $\mathbb{E}[(X_k - \mathbb{E}[X_k])^i]$ is the i -th central moment of X_k .

The first step is to **find the Taylor expansion** of $L_n(f)(\lambda)$. Given that $L_n((e_1 - x)^0)(x) = L_n((e_1 - x)^1)(x) = 0$, this becomes:

$$L_n(f)(\lambda) = L_n(R_3(f, \lambda)) + \sum_{i=2}^3 L_n((e_1 - \lambda)^i)(\lambda) \frac{f^{(i)}(\lambda)}{i!},$$

$$\text{abs}(L_n(f)(\lambda)) \leq \|L_n(R_3(f, \lambda))\| + \|L_n((e_1 - \lambda)^2)\| \cdot \|f^{(2)}\|/2$$

$$+ \|L_n((e_1 - \lambda)^3)\| \cdot \|f^{(3)}\|/6.$$

The function $\text{abs}(L_n((e_1 - x)^3)(x))$ has its maximum at $x = 1/2 - \sqrt{3}/6$; and $\text{abs}(L_n((e_1 - x)^2)(x))$ has its maximum at $x = 1/2$, so:

$$\text{abs}(L_n(f)(\lambda)) \leq \|L_n(R_3(f, \lambda))\| + \text{abs}\left(\frac{3\lambda(\lambda - 1)}{2n(2n - 1)}\right) \|f^{(2)}\|/2$$

$$+ \|L_n((e_1 - \lambda)^3)\| \cdot \|f^{(3)}\|/6$$

$$\leq \|L_n(R_3(f, \lambda))\| + \frac{3}{8n(2n - 1)} \|f^{(2)}\|/2$$

$$+ \frac{\sqrt{3}(6n - 5)}{24n^2(2n - 1)} \|f^{(3)}\|/6.$$

Meanwhile the remainder is estimated as follows, using the proof of corollary 2.3 of Gonska et al. (2006)⁷⁸:

$$\|L_n(R(f, \lambda))\| \leq \frac{1}{6} \|f^{(3)}\| \cdot \|(LA_n + LB_n)(\text{abs}(e_1 - \lambda)^3)\|.$$

⁷⁶Wang, Y.Q., Zhang, Y.Y, Liu, J.L., “Expectation identity of the hypergeometric distribution and its application in the calculations of high-order origin moments”, Communications in Statistics–Theory and Methods 52(17), 2023. <https://doi.org/10.1080/03610926.2021.2024235>

⁷⁷Weisstein, Eric W. “Central Moment.” From MathWorld–A Wolfram Resource. <https://mathworld.wolfram.com/CentralMoment.html>

⁷⁸Gonska, Heiner, Paula Pi ul, and Ioan Raşa. “On differences of positive linear operators.” Carpathian Journal of Mathematics (2006): 65-78.

In turn, using Schwarz's inequality (see proof of the same paper's corollary 2.1):

$$\begin{aligned} \| (LA_n + LB_n)(\text{abs}(e_1 - \lambda)^3) \| &\leq (\| (LA_n + LB_n)((e_1 - \lambda)^4) \|)^{1/2} \\ &\times (\| (LA_n + LB_n)((e_1 - \lambda)^2) \|)^{1/2} \leq \frac{3\sqrt{3}}{8n^{3/2}}. \end{aligned}$$

(The expression in the middle takes its maximum at $\lambda = 1/2$; the right-hand side is an upper bound of that expression for all positive integers n .) Altogether:

$$\begin{aligned} \|L_n(f)\| &\leq \frac{3}{8n(2n-1)} \frac{1}{2} \|f^{(2)}\| \\ &+ \left(\frac{3\sqrt{3}}{8n^{3/2}} + \frac{\sqrt{3}(6n-5)}{24n^2(2n-1)} \right) \frac{1}{6} \|f^{(3)}\| = LC_n(f) \\ &\leq 0.1875 \frac{\|f^{(2)}\|}{n^{3/2}} + \frac{5\sqrt{3}}{72} \frac{\|f^{(3)}\|}{n^{3/2}} \leq \frac{0.3078M}{n^{3/2}} = O(1/n^{3/2}). \end{aligned}$$

If $n \geq 2$ is an integer, $LC_n(f) \leq 0.2165M/n^{3/2}$.

9 Example: An Interesting Linear Operator

For a continuous function f on the closed unit interval and for nonnegative integers m and n , let $H_{n,m}$ be a linear operator as follows:

$$H_{n,m}(f) = B_n(f) + \text{Lag}_m(f) - B_n(\text{Lag}_m(f)),$$

where B_n is the degree- n Bernstein polynomial (see “Bernstein Form and Bernstein Polynomials”, earlier) and Lag_m is the polynomial of degree up to m that equals f at “ $m+1$ distinct points on” the closed unit interval. This operator was mentioned in Remark 2 of Gavrea and Ivan (2018)⁷⁹, but appears not to have been studied elsewhere.

It is known that Lag_m is a linear operator and reproduces all polynomials of degree m or less, so that $\text{Lag}_m(e_i) = e_i$ whenever $0 \leq i \leq m$ is an integer. Thus, if f is such a polynomial, $B_n(f) = B_n(\text{Lag}_m(f))$ and therefore $H_{n,m}(f) = \text{Lag}_m(f) = f$, and therefore $H_{n,m}(e_i) = e_i$ whenever $0 \leq i \leq m$ is an integer.

(The foregoing sentence would remain true if B_n were replaced with any other operator mapping to and from the same functions.)

Because $H_{n,m}$ is linear and reproduces all polynomials up to degree m , the following holds if f has a continuous m -th derivative:

$$\begin{aligned} H_{n,m}(f)(\lambda) - f(\lambda) &= H_{n,m}(R_m(f, \lambda))(\lambda) \\ &= B_n(R_m(f, \lambda)) + \text{Lag}_m(R_m(f, \lambda)) - B_n(\text{Lag}_m(R_m(f, \lambda))). \end{aligned}$$

⁷⁹Ioan Gavrea, Mircea Ivan, “A note on the fixed points of positive linear operators”, Journal of Approximation Theory (227), 2018, <https://doi.org/10.1016/j.jat.2017.12.001>.

With the help of Lemma 6, the following holds if n is also 2 or greater and m is a positive integer:

$$\|H_{n,m}(f)(\lambda)\| \leq \frac{\|f^{(m)}\|\mu_m}{(m!)n^{m/2}} + \|\text{Lag}_m(R_m(f, \lambda)) - B_n(\text{Lag}_m(R_m(f, \lambda)))\|$$

where μ_r is as in Proposition 1.

It will be helpful to estimate the second derivative of $\text{Lag}_m(f)$.

Given that that function is a polynomial of degree m or less, this can be estimated as:

$$\text{abs}(\text{Lag}_m(f)^{(2)}(\lambda)) \leq \|\text{Lag}_m\| \cdot \|f\| \cdot M(m), \quad (\text{E})$$

where:

- $\|\text{Lag}_m\|$ is the operator norm of Lag_m , which in this case equals its *Lebesgue constant*, which will vary depending on the points on the closed unit interval where the polynomial meets (interpolates) f .
- $M(m) = (4/3) \cdot \max(2, m)^2(\max(2, m)^2 - 1)$. This is an upper bound on the maximum absolute value of a polynomial's second derivative (on the closed unit interval) when that polynomial has a maximum absolute value of 1 (on that interval). This uses the following lemma based on one proved for the interval $[-1, 1]$ by V. Markov in 1892 (see also Schaeffer and Duffin 1938⁸⁰).

Lemma: Let $p(\lambda) = c_0\lambda^0 + \dots + c_n\lambda^n$ be a polynomial on the interval $[a, b]$, where $c_0, \dots, c_n$ are real numbers and c_n is not zero. If $\|p\| \leq 1$, then $\|p^{(2)}\| \leq (b-a)^2 n^2(n^2-1)/3$.

An error bound for $H_{n,m}$ can also be written as:

$$\begin{aligned} H_{n,m}(f) - f &= B_n(f) + \text{Lag}_m(f) - B_n(\text{Lag}_m(f)) - f \\ &= (B_n(f) - f) + (\text{Lag}_m(f) - B_n(\text{Lag}_m(f))), \end{aligned}$$

$$\text{abs}((H_{n,m}(f) - f)(\lambda)) \leq \text{abs}(B_n(f) - f) + \text{abs}(B_n(\text{Lag}_m(f)) - \text{Lag}_m(f)),$$

so now there are two error bounds to find: one for f and the other for $\text{Lag}_m(f)$. And, if f has a continuous second derivative, both have the same form:

$$B_n(g) \leq \|f^{(2)}\|/(8n).$$

(This follows from Lorentz (1963)⁸¹ and the well-known fact that $\|g^{(2)}\|$, the maximum absolute value of g 's second derivative, is an upper bound of g 's first derivative's smallest Lipschitz constant.)

Altogether, if f has a continuous second derivative and m is fixed:

$$\text{abs}((H_{n,m}(f) - f)(\lambda)) \leq \frac{\|f^{(2)}\|}{8n} + \frac{\|\text{Lag}_m\| \cdot \|f\| \cdot M(m)}{8n}.$$

It is suspected that, using (E), the following is true if f has a continuous m -th derivative and m is a positive integer:

⁸⁰Schaeffer, A. C., and R. J. Duffin. "On some inequalities of S. Bernstein and W. Markoff for derivatives of polynomials." Bulletin of the American Mathematical Society 44.4 (1938): 289-297.

⁸¹G.G. Lorentz, "Inequalities and saturation classes for Bernstein polynomials", 1963.

$$H_{n,m}(f)(\lambda) - f(\lambda) = B_n(R_m(f, \lambda)) + (\text{Lag}_m(f) - B_n(\text{Lag}_m(f))),$$

so that, using Lemma 6 again, there is a better error bound:

$$\text{abs}((H_{n,m}(f) - f)(\lambda)) \leq \frac{\|f^{(m)}\| \mu_m}{(m!)n^{m/2}} + (\|Lag_m\| \cdot \|f\| \cdot M(m)) \frac{\mu_m}{(m!)n^{m/2}}.$$

10 Example: The Lorentz Operators

The *Lorentz operators* were introduced by Lorentz (1963)⁸² and studied by Holtz et al. (2011)⁸³.

This section touches on the Lorentz operator of order 2, defined as—

$$\begin{aligned} Q_{n,2}(f)(\lambda) &= B_n(f)(\lambda) - \frac{\lambda(1-\lambda)}{2n} B_n(f^{(2)})(\lambda) \\ &= B_n \left(f(e_1) - \frac{\lambda(1-\lambda)}{2n} f^{(2)}(e_1) \right) (\lambda), \end{aligned}$$

where $0 \leq \lambda \leq 1$.⁸⁴

This operator is a nonpositive linear operator that maps continuous functions to polynomials of degree $n+2$. Because $Q_{n,2}(e_i) = e_i$ if i is 0, 1, or 2, the operator reproduces all polynomials of degree 2 or less (for another proof, see Lemma 14 of Holtz et al. 2011⁸⁵). (The Lorentz operators of order 0 and 1 are simply the Bernstein polynomials.)

$Q_{n,2}$ can be bounded as follows:

$$\|Q_{n,2}(f)\| \leq \|B\| + \frac{1}{8n} \|B\| \cdot \|f^{(2)}\| = 1 + \|f^{(2)}\|/(8n),$$

where $\|B\|$ is the operator norm of the Bernstein polynomials.

However, this operator has an infinite operator norm and so is unbounded. Indeed, a function can have a maximum absolute value of 1 or less yet have an arbitrarily large and continuous second derivative. Take, for example, the family of functions $g_m(\lambda) = \cos(m \cos(\lambda\pi))$: as m goes to infinity, $g_m^{(2)}(1/2)$ diverges; in turn, the expression $(B_n(g_m^{(2)})(1/2)) \frac{1}{8n}$, based on a degree- n Bernstein polynomial, likewise diverges, at least if n is 7 or less.

Because $Q_{n,2}$ is unbounded, lemmas 7 and 8 (see “**Bounds for General Linear Operators**”), both of which relate to Peano kernels and apply only to bounded operators, can’t be used. For the same reason, $Q_{n,2}$ cannot be written as a difference of two positive linear operators that map continuous functions to continuous functions; every positive operator of that kind is bounded (Pițul 2007, Corollary 1.5)⁸⁶.

⁸²Lorentz, G.G. The degree of approximation by polynomials with positive coefficients. Math. Ann. 151, 239–251 (1963). <https://doi.org/10.1007/BF01398235>

⁸³Holtz, O., Nazarov, F. & Peres, Y. New Coins from Old, Smoothly. Constr Approx 33, 331–363 (2011). <https://doi.org/10.1007/s00365-010-9108-5>

⁸⁴ $Q_{n,2}$ can also be seen as the Bernstein polynomial of a so-called *linear differential operator*: $1 \cdot f^{(0)} + 0 \cdot f^{(1)} + (\lambda(1-\lambda)/(2n)) \cdot f^{(2)}$.

⁸⁵Holtz, O., Nazarov, F. & Peres, Y. New Coins from Old, Smoothly. Constr Approx 33, 331–363 (2011). <https://doi.org/10.1007/s00365-010-9108-5>

⁸⁶Pițul, P., “Evaluation of the Approximation Order by Positive Linear Operators”, dissertation, Universität Duisberg-Essen, 2007

Some of the “moments” of this operator are:

- $Q_{n,2}(e_3) = \frac{x(n^2x^2+2x^2-3x+1)}{n^2}$.
- $Q_{n,2}((e_1 - \lambda)^3) = \frac{x(2x^2-3x+1)}{n^2}$.

Let $LF = f - Q_{n,2}(f)$ be the error in approximating f with $Q_{n,2}(f)$.

It is suspected that—

$$\|LF(f)\| \leq \frac{C\|f^{(3)}\|}{n^{3/2}},$$

for some $C > 0$, and it is of interest to find an explicit upper bound for C , especially a tight one.

11 Probabilistic Interpretations of Linear Operators

The Bernstein polynomials featured in a proof in 1912 of the result that any continuous function on a compact interval can be approximated as well as desired by polynomials (Bernstein 1912)⁸⁷. That proof used probability theory. In a series of papers, Adell and De la Cal use probability theory to interpret a number of linear operators in addition to those polynomials (Adell and De la Cal 1996⁸⁸, 1995⁸⁹).

12 Conclusion and Ways to Improve This Article

Many academic papers and books present results in approximation theory, especially error bounds, with hidden constants and without giving upper bounds for such constants. For example, a result may say the following about an operator (approximation scheme) L and a function $f(\lambda)$. If certain conditions are met, then:

$$\text{abs}(f(\lambda) - L(\lambda)) \leq C\omega_1(f, 1/n^{1/2}), \text{ or}$$

$$\text{abs}(f(\lambda) - L(\lambda)) \leq D\|f^{(3)}\|/n^{3/2},$$

where C and D are unspecified constants with no upper bounds given. Or:

$$\text{abs}(f(\lambda) - L(\lambda)) = O(1/n^{1/2}),$$

where $O(1/n^{1/2})$ is a function whose absolute value is no more than an unspecified constant times $1/n^{1/2}$. (For example, compare Sevy 1991⁹⁰ with Gonska and Zhou 1994⁹¹ and Holtz et al. 2011⁹².)

⁸⁷S.N. Bernstein, “Démonstration du théorème de Weierstrass fondée sur le calcul des probabilités”, Comm. Kharkov Math. Soc. 13, 1-2, 1912.

⁸⁸Adell, J. A., and J. De la Cal. “Bernstein-type operators diminish the ω -variation.” Constructive Approximation 12.4 (1996): 489-507. <https://doi.org/10.1007/BF02437505>

⁸⁹Adell, J. A., and J. De la Cal. “Bernstein-Durrmeyer operators.” Computers & Mathematics with Applications 30.3-6 (1995): 1-14. <https://doi.org/10.1016/0898-1221%2895%2900081-X>

⁹⁰Sevy, J., “Acceleration of convergence of sequences of simultaneous approximants”, dissertation, Drexel University, 1991. <https://doi.org/10.17918/00010296>

⁹¹Gonska, Heinz H., and Xin-long Zhou. “Approximation theorems for the iterated Boolean sums of Bernstein operators.” Journal of Computational and Applied Mathematics 53.1 (1994): 21-31. [<https://doi.org/10.1016/0377-0427%02892%02900133-T>]

⁹²Holtz, O., Nazarov, F. & Peres, Y. New Coins from Old, Smoothly. Constr Approx 33, 331–363 (2011). <https://doi.org/10.1007/s00365-010-9108-5>

It was a goal of this article to catalog general-purpose error bounds without such hidden constants.

To improve this article, explicit error bounds (with no hidden constants) of the following kinds are sought:

- Inequalities similar to the **Lebesgue inequality** for certain operators where the normal Lebesgue inequality doesn't apply (say, spline operators with local approximation, or numerical integration rules).
- General-purpose upper bounds on the error when approximating a function with:
 1. Polynomials, especially polynomials in Bernstein form with nonnegative coefficients.
 2. Ratios of polynomials described in (1).
 3. Convex combinations of functions described in (1) or (2). A convex combination has the form $c_0 f_0(\lambda) + c_1 f_1(\lambda) + \dots$ where c_i are nonnegative and sum to 1.
 4. Compositions of functions described in (1), (2), or (3). A composition of functions f and g is a function like $f(g(\lambda))$.
- Easy-to-use upper bounds for **estimating Peano kernels**, such as the Peano kernels corresponding to the **Lorentz operators**.
- Additional error bounds for **nonlinear operators**.
- Error bounds that are sharper than those in Lemmas 9 to 12 for functions with a k -th continuous derivative for some $k \geq 2$.

In addition, the following will be helpful.

- Results on ways to rewrite a nonpositive linear operator into a difference of two positive linear operators, as used in Lemma 8 and the first Example section.
- A cleaner way to prove a linear operator is unbounded. For example, the Lorentz operator is unbounded largely because it requires the values of second derivatives.
- Does Lemma 7 also work for unbounded linear operators, such as the **Lorentz operator**?

Also:

- Is it helpful to introduce the concept of functions of bounded variation in this article?

Those are not all the possible ways this article can be improved.

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14 Notes

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